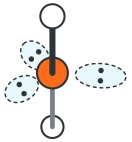
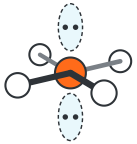
VSEPR Shapes

CHEMISTRY REFERENCE

	0 lone pairs	1 lone pair	2 lone pairs	3 lone pairs
2 ELECTRON DOMAINS ELECTRON GEOMETRY **Linear**	 **Linear** 2 bonding · 0 lone | 180° | AX_2			
3 ELECTRON DOMAINS ELECTRON GEOMETRY **Trigonal Planar**	 **Trigonal Planar** 3 bonding · 0 lone | 120° | AX_3	 **Bent** 2 bonding · 1 lone | <120° | AX_2E_1		
4 ELECTRON DOMAINS ELECTRON GEOMETRY **Tetrahedral**	 **Tetrahedral** 4 bonding · 0 lone | 109.5° | AX_4	 **Trigonal Pyramidal** 3 bonding · 1 lone | ≈107° | AX_3E_1	 **Bent** 2 bonding · 2 lone | ≈104.5° | AX_2E_2	
5 ELECTRON DOMAINS ELECTRON GEOMETRY **Trigonal Bipyramidal**	 **Trigonal Bipyramidal** 5 bonding · 0 lone | 90°, 120° | AX_5	 **Seesaw** 4 bonding · 1 lone | <90°, <120° | AX_4E_1	 **T-Shaped** 3 bonding · 2 lone | <90° | AX_3E_2	 **Linear** 2 bonding · 3 lone | 180° | AX_2E_3
6 ELECTRON DOMAINS ELECTRON GEOMETRY **Octahedral**	 **Octahedral** 6 bonding · 0 lone | 90° | AX_6	 **Square Pyramidal** 5 bonding · 1 lone | <90° | AX_5E_1	 **Square Planar** 4 bonding · 2 lone | 90° | AX_4E_2	

central atom (A) ○ bonded atom (X) ◌ lone pair (E) grey bond = pointing behind the page

Electron domains = bonded atoms + lone pairs on the central atom. A double or triple bond counts as one domain.